MINERAL

A mineral is homogenous, naturally occurring, inorganic solid substance formed through geological processes that has a characteristic chemical composition, a highly ordered atomic structure and specific physical properties

Different Forms/habits of minerals

Chemical Composition and Internal Structure of Minerals

- Elements are the building blocks of minerals.
- Some minerals exist as single elements; however, most minerals consist of a combination of several elements joined by a chemical bond to form a stable mineral compound.
- Elements chemically bond to one another when their atoms gain, lose, or share electrons with other elements.
- In addition to ionic and covalent bonds, other bonds can also occur through various combinations of transferred and shared electrons.
- Nearly 4,000 minerals are identified on the planet Earth, and new minerals continue to be discovered all the time.

Geochemical Classification of elements

- **Lithophile** elements are preferentially partitioned into silicate minerals. These include cations that commonly form oxides, such as Ca, Mg, Mn, Ti, Na, K, U, Th, Si, and Fe in its oxidized states. Lithophile elements also occur naturally as oxides, halides, phosphates, sulfates, and carbonates and are concentrated in the silicate portion (i.e. crust and mantle) of the earth.
- **Chalcophile** elements are those metals and heavier non-metals that have a low affinity for oxygen and prefer to bond with sulphur to form sulphide type minerals or highly insoluble sulphides. These include Cu, Pb, Zn, Cd, Mo, Hg, Sb, Sn, Tl, Te, As etc.
- **Siderophile** elements ("iron loving") are those that are preferentially partitioned into the metallic core, typically in the form of alloys with Fe. They are depleted in the silicate portion of the earth and presumably concentrated in the core. Sulfur may dissolve in the core as a sulfide complex and, under these conditions, is also considered siderophile.
- **Atmophile** elements are those that readily form volatile (e (i.e., they form gases or liquids at the surface of the Earth) compounds at relatively low temperatures

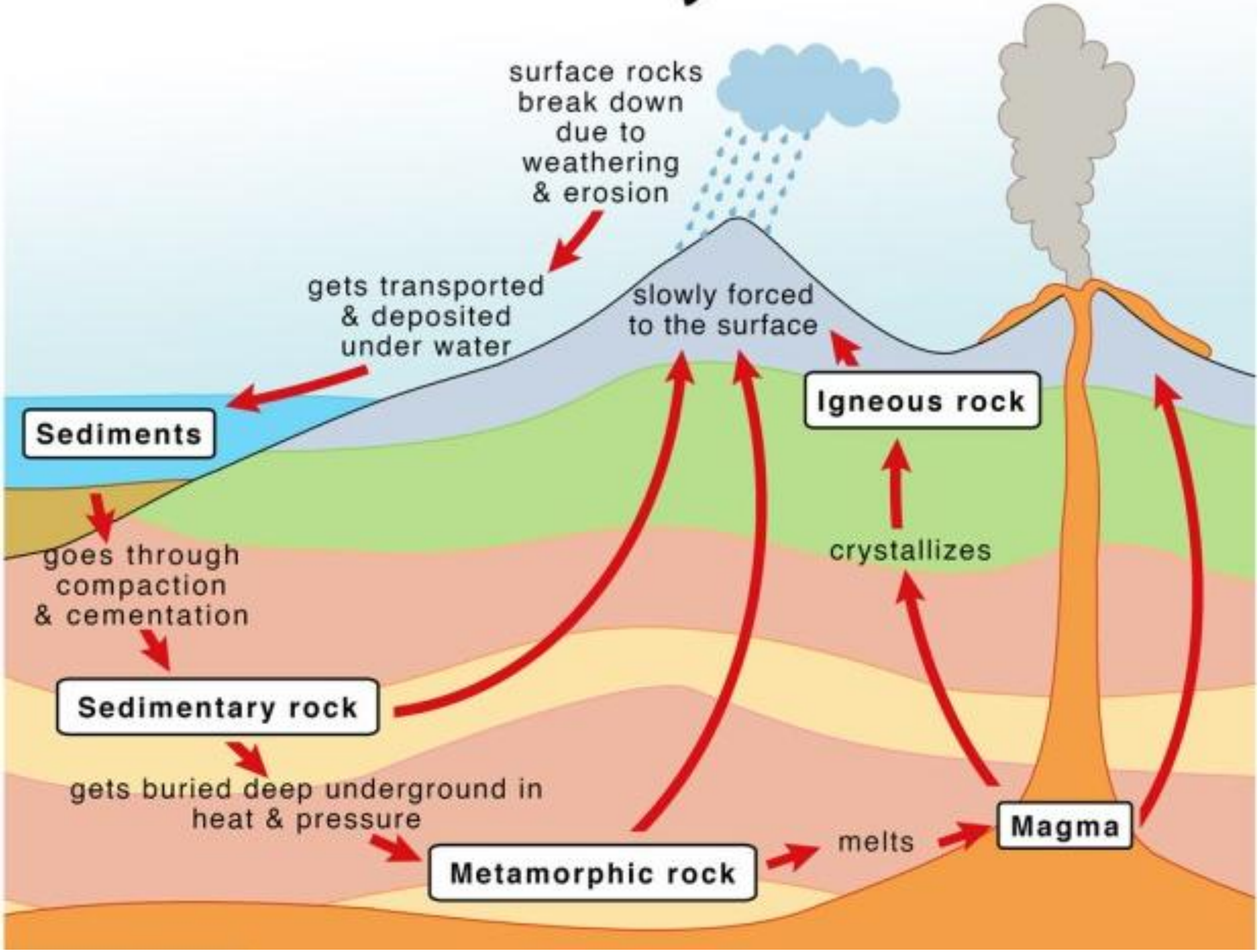
Geochemical Classification of Elements

Siderophile	Chalcophile	Lithophile	Atmophile
Fe^{*} Co^{*} Ni^{*}	(Cu) Ag	Li Na K Rb Cs	(H) (C) N (O)
Ru Rh Pd	Zn Cd Hg	Be Mg Ca Sr Ba	(Cl) (Br) (I)
Os Ir Pt	Ga In Tl	B Al Sc Y REE	He Ne Ar
Au Re^{+} Mo^{+}	(Ge) (Sn) Pb	Si Ti Zr Hf Th	Kr Xe
Ge^{*} Sn^{*} W^{++}	(As) (Sb) Bi	P V Nb Ta	
C^{++} Cu^{*} Ga^{*}	S Se Te	O Cr U	
(P) As^{+} Sb^{+}	(Fe) Mo (Os)	H F Cl Br I	
	(Ru) (Rh) (Pd)	(Fe) Mn (Zn) (Ga)	

ROCKS

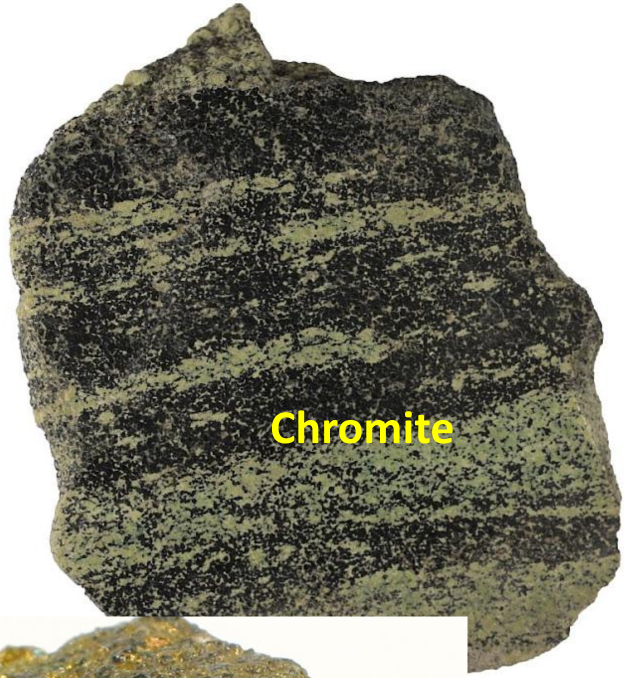
- Rocks are naturally occurring solid aggregates or masses of minerals, mineraloids, or organic material that make up the Earth’s crust. They are composed of one or more minerals in different proportions , which are crystalline solids with a specific chemical composition and a defined atomic structure.
- Rocks can vary greatly in size, shape, colour, texture, and composition, and they are classified into three main types based on their formation process: **igneous, sedimentary, and metamorphic rocks.**

Rock Cycle


surface rocks break down due to weathering & erosion
gets transported & deposited under water
slowly forced to the surface
Sediments
Igneous rock
goes through compaction & cementation
crystallizes
Sedimentary rock
gets buried deep underground in heat & pressure
Magma
melts
Metamorphic rock

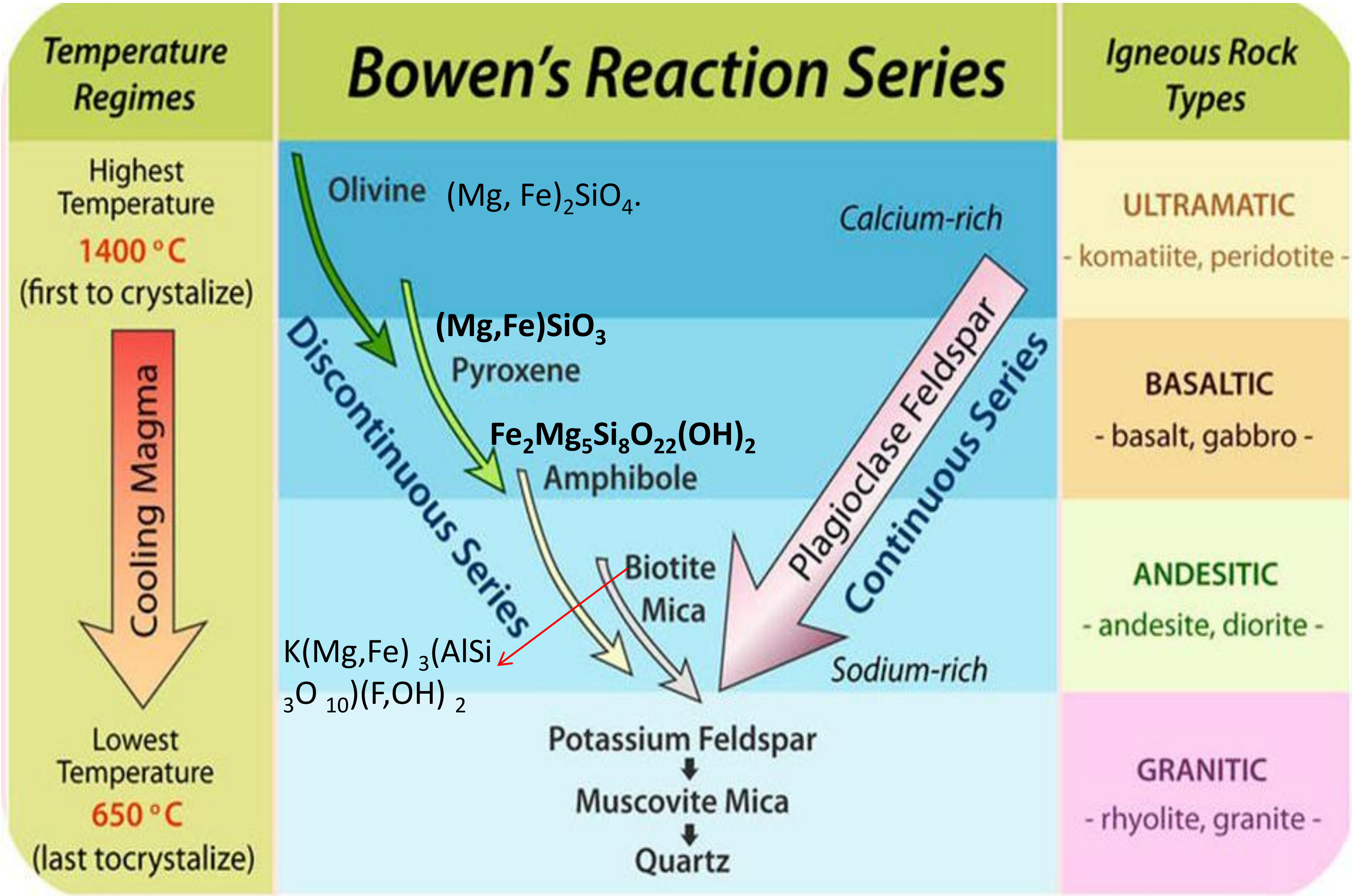
Mineral-Rock Relationship

Host rock and Ore Association

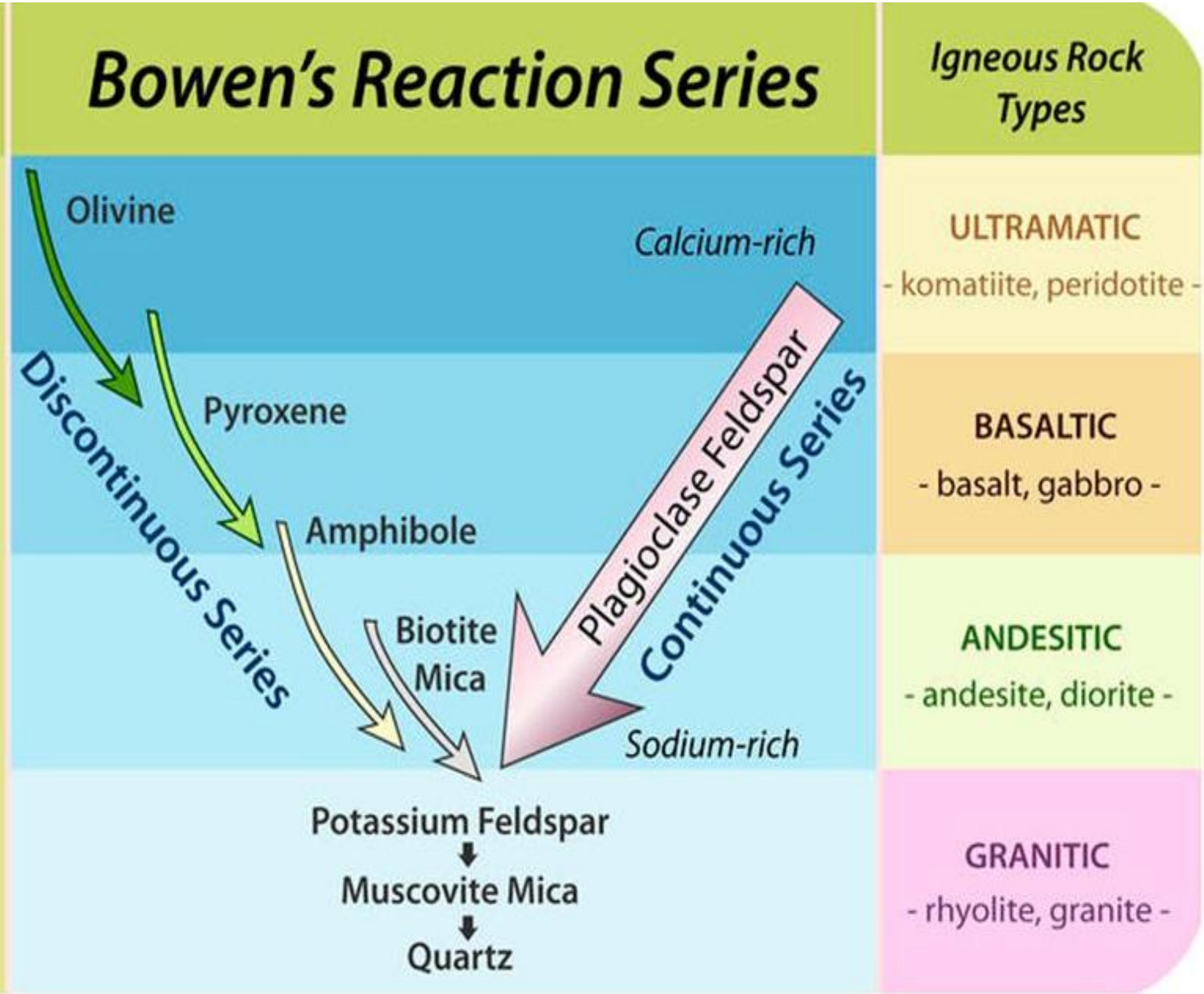


How Do Minerals Grow?

- New minerals are forming everyday on the Earth's surface, in the Earth's crust, and deep within the Earth's interior.
- Minerals form from molten rock and volcanic magma within the Earth's interior and crust. In these environments, changes in temperature and pressure and chemical composition influence the type of minerals which form, the size of their individual crystals, and their growth rate.
- Minerals grow from saturated solutions in rock cavities. Differences in temperature, chemical composition, and the saturation content of the solution influence the type of minerals which form, the size of their individual crystals, and their growth rate.
- The arrangement of atoms during crystal formation determines what the mineral will be and what crystal shape it will have.
- The crystal form is one of several characteristics that Geologists use to identify different minerals.



Temperature Regimes
Bowen's Reaction Series
Igneous Rock Types
Highest Temperature
1400 °C
(first to crystalize)
Cooling Magma
Lowest Temperature
650 °C
(last tocrystalize)
Olivine $(Mg, Fe)_2SiO_4$.
$(Mg,Fe)SiO_3$
Pyroxene
$Fe_2Mg_5Si_8O_{22}(OH)_2$
Amphibole
Biotite Mica
$K(Mg,Fe)_3(AlSi_3O_{10})(F,OH)_2$
Discontinuous Series
Calcium-rich
Plagioclase Feldspar
Continuous Series
Sodium-rich
Potassium Feldspar
Muscovite Mica
Quartz
ULTRAMATIC
- komatiite, peridotite -
BASALTIC
- basalt, gabbro -
ANDESITIC
- andesite, diorite -
GRANITIC
- rhyolite, granite -



Chromite, PGE, Ni, V, Ti, Co

Ni, Ti, Cu, Au, Ag, Zn

Cu, Sn, Nb, W, Mo

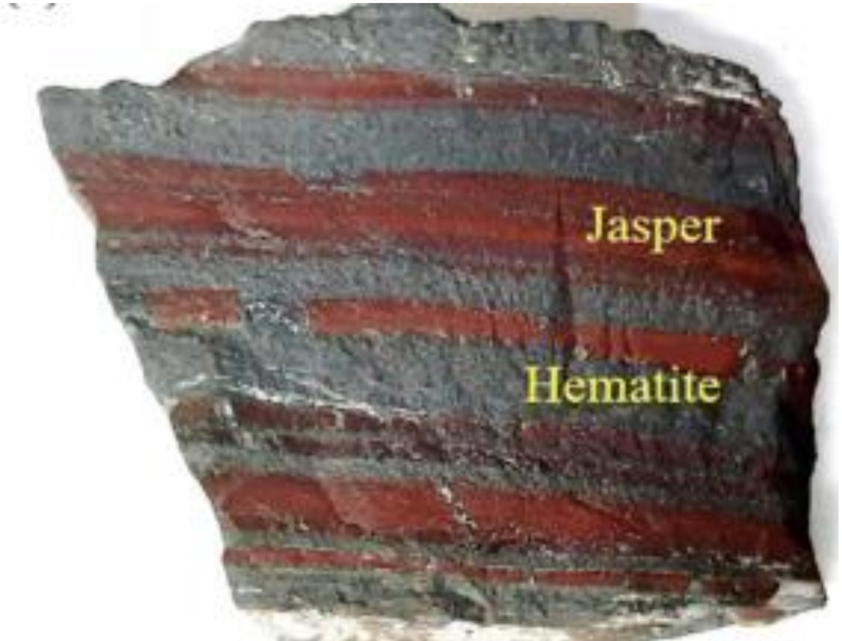
REE, Li

Economic Geology

Economic geology deals with metal ores, fossil fuels (e.g., petroleum, natural gas, and coal), and other materials of commercial value, such as salt, gypsum, and building stone.

A mineral deposit (or an ore deposit) may be defined as a rock body that contains one or more elements (or minerals) minerals in an unusually high concentration to have potential economic value.

Hematitic Ore (Khondbond, Odisha)



The mineral deposit can be broadly categorised into

1) metallic ore mineral deposit from which one or more metal can be extracted and
2) non-metallic or industrial mineral deposits.

- Geologists classify mineral deposits in many different ways, according to the:
- 1) Commodity being mined
- 2) Tectonic setting in which the deposit occurs
- 3) Geological setting of the mineral deposit
- 4) Genetic model for the origin of the ore deposit

Classification of metallic and non-metallic minerals

- <u>Metallic Minerals</u>
- Ferrous Metals (Fe, Ti, Cr, Mn etc.)
- Non-Ferrous Metals (Cu, Pb, Zn, Sb, Ni etc.)
- Noble Metals (Au, Ag, Pt, Pd, Rh, Os, Ir)
- Light Metals (Al, Li, Be, Mg etc)
- Rare earths and dispersed Metals (La, Ce to Lu) and (Sc, Ga, Rb, Cd etc)
- Rare and Scarce Metals (W, Mo, Sn, Co, Hg, Bi, Zr, Cs, Nb, Ta etc.)

Non Metallic Materials

1. Fossil Fuels (Coal, Petroleum and natural gas)
2. Structural and building material (Granite, Khondalite, Dolerite, Slate, Gypsum, etc)
3. Ceramic materials (Clays, Feldspar, Bauxite, Magnesite, Barite etc)
4. Metallurgical and Refractory materials (Fluorspar, Graphite, Magnesite, dolomite, Bauxite, Zircon etc)
5. Industrial and manufacturing materials (Asbestos, Mica, Talc, Barite etc)
6. Chemical materials (Salt, Borates, Ca,Mg,Cl_2, Bromine, Nitrates etc)
7. Abrasives and abrasive materials (Corundum, Emery, Garnet etc)

Classification of Minerals

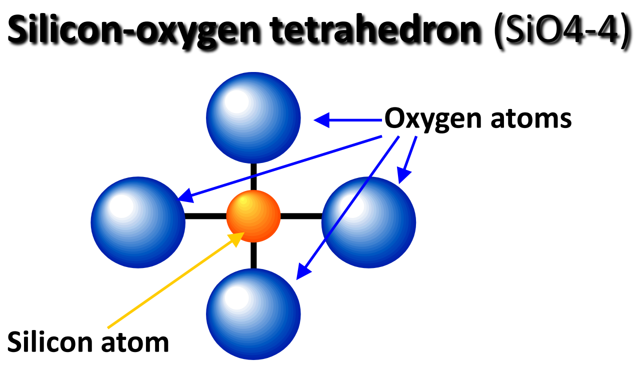
Minerals are classified by their chemical composition and internal crystal structure.

There are 7 Major Mineral Groups:

- Silicates
- Native Elements
- Halides
- Carbonates
- Oxides
- Sulfates
- Sulfides

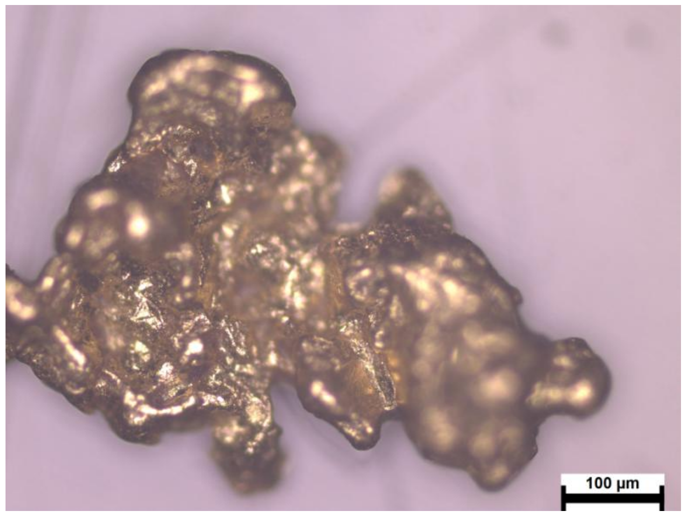
Silicates

- Silicates are composed of silicon-oxygen tetrahedrons, an arrangement which contains four oxygen atoms surrounding a silicon atom (SiO_4^{-4}).
- Silicates are often divided into two major groups: ferromagnesian silicates and non-ferromagnesian silicates
  - Ferromagnesian silicates contain iron or magnesium ions joined to the silicate structure. They are darker and have a heavier specific gravity than non-ferromagnesian silicate minerals.
  - Ferromagnesians include minerals such as olivine, pyroxene, hornblende, and biotite
  - Non-ferromagnesians include muscovite, feldspar, and quartz
- Silicates comprise the majority of minerals in the Earth's crust and upper mantle. Over 25% of all minerals are included in this group, with over 40% of those accounting for the most common and abundant minerals.
- Feldspar, Quartz, Biotite, and Amphibole are the most common silicates

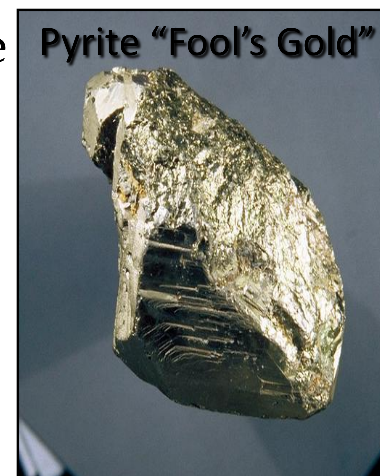


Native Elements

- Native elements are minerals that are composed of a single element.
- Some examples are: Gold (Au), Silver (Ag), Copper (Cu), Iron (Fe), Diamonds (C), Graphite (C), and Platinum (Pt)



- Sulfides are minerals composed of one or more metal cations combined with sulfur. Many sulfides are economically important ores.
- Pyrite (FeS_2) or “fool’s gold”, Galena (PbS), Sphalerite (Zns), Cinnabar (HgS) an Molybdenite (MoS_2) are a few commonly occurring sulfide minerals



Cinnabar

Halides

- Halides consist of halogen elements, chlorine (Cl), bromine (Br), fluorine (F), and iodine (I) forming strong ionic bonds with alkali and alkali earth elements sodium (Na), calcium (Ca) and potassium (K)

- Some examples include Halite (NaCl) and Flourite (CaF_2).

Halite

Fluorite

Oxides

- Oxides are minerals that include one or more metal cations bonded to oxygen or hydroxyl anions.
- Examples of oxide minerals include: Hematite (Fe_2O_3), Magnetite (Fe_3O_4), Corundum (Al_2O_3), and Ice (H_2O)

Carbonates

- Carbonates are anionic groups of carbon and oxygen. Carbonate minerals result from bonds between these complexes and alkali earth and some transitional metals
- Common carbonate minerals include calcite $CaCO_3$, calcium carbonate, and dolomite $CaMg(CO_3)_2$, calcium/magnesium carbonate
- Carbonate minerals react when exposed to hydrochloric acid. Geologist will often carry dilute hydrochloric acid in the field to test if a mineral contains calcium carbonate. If the mineral fizzes when it comes in contact with the hydrochloric acid it contains calcium carbonate. Some cola soft drinks can also be used for this test because it contains enough hydrochloric acid to react with calcium carbonate.

Dolomite

Calcite

Sulfates

- Sulfates are minerals that include SO_4 anionic groups combined with alkali earth and metal cations.
- Anhydrous (no water) and hydrous (water) are the two major groups of Sulfates.
- Barite ($BaSO_4$) is an example of a anhydrous sulfate and Gypsum ($CaSO_4 \cdot 2H_2O$) is an example of a sulfate.

Mineral Properties

Minerals have distinctive physical properties that geologists use to identify and describe them. There are 7 major physical properties of minerals:

- 1. Crystal Form
- 2. Hardness
- 3. Luster
- 4. Color
- 5. Streak
- 6. Cleavage
- 7. Specific Gravity

Special Properties

Magnetism

Acid reaction

Odour

Taste

Hardness

- Hardness is the ability of a mineral to resist abrasion or scratching on its surface.
- One way geologists measure hardness is using a relative scale referred to as Moh's scale of mineral hardness which ranks 10 common minerals along a scale from 1-10 (1 refers to the softest minerals while 10 refers to the hardest mineral).
- Geologists measure a mineral's hardness by scratching the surface of a mineral using minerals of known hardness, or by scratching the surface using a variety of other hardness indicators such as fingernails, pennies, or glass.

Talc

Talc is a soft mineral that you can scratch with your fingernail, and has a hardness of "1" measured by Moh's relative scale of mineral hardness.

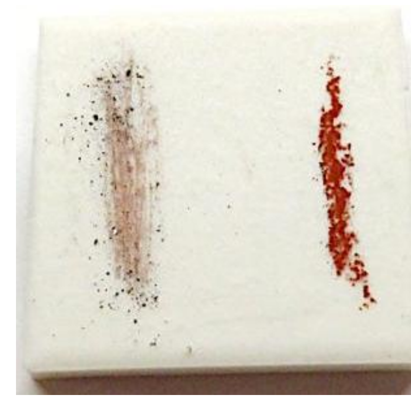
Mohs scale of hardness

Mohs hardness	Mineral	Chemical formula	Absolute hardness
1	Talc	$Mg_3Si_4O_{10}(OH)_2$	1
2	Gypsum	$CaSO_4{\cdot}2H_2O$	3
3	Calcite	$CaCO_3$	9
4	Fluorite	CaF_2	21
5	Apatite	$Ca_5(PO_4)_3(OH^-,Cl^-,F^-)$	48
6	Orthoclase Feldspar	$KAlSi_3O_8$	72
7	Quartz	SiO_2	100
8	Topaz	$Al_2SiO_4(OH^-,F^-)_2$	200
9	Corundum	Al_2O_3	400
10	Diamond	C	1600

Approximating Hardness

- Take the unknown mineral and attempt to scratch with your fingernail (H=2.5), copper penny (H=3.5), a glass plate (H=5.5), and a streak plate (H=6.5).

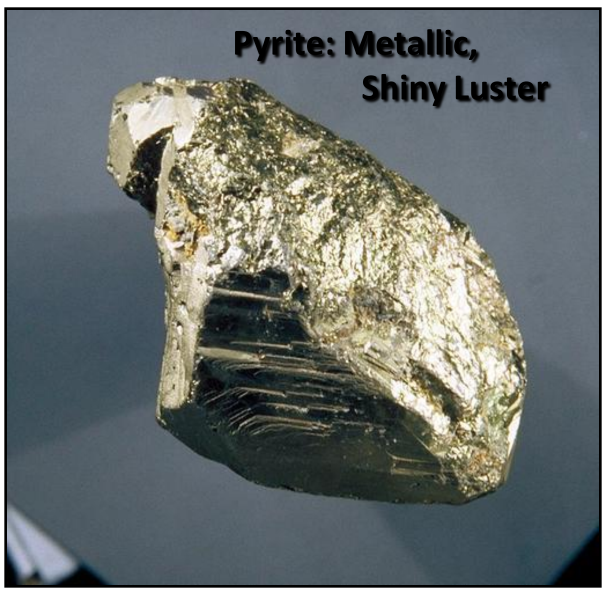
- If the mineral scratches any of the materials, then it is harder than that material.



- If it scratches your fingernail and not the penny, than the hardness is between 2.5 and 3.5, probably 3.0.
- By this process, we can determine the approximate hardness of the unknown mineral.
- We do not need to know the exact hardness of the mineral because we will use other physical properties to refine the identification.

Luster

- Luster refers to how light is reflected from the surface of a mineral.
- There are two main types of luster: metallic and non-metallic:
  - Minerals with a metallic luster are described as shiny, silvery, or having a metal-like reflectance.
  - Non-metallic minerals may be described as resinous, translucent, pearly, waxy, greasy, silky, vitreous/glassy, dull, or earthy
- Luster may be subjective, and thus is not always a reliable identifier

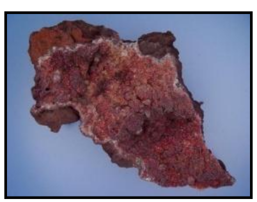


Color

- Mineral color is determined by how the crystals absorb and reflect light. Although color is easy to recognize, it is often misleading.
- Minerals, such as quartz, fluorite, halite, and calcite occur in a wide variety of colors, and other minerals, such as olivine, malachite, and amphibole have fairly distinctive colors.
- Variations in a mineral's color may be the result of impurities in the atomic structure of the crystal or the presence of a particular chemical when the crystal formed.
- Because some minerals can occur in several colors, color is generally not a good characteristic for describing and identifying minerals.

Different Colors of Calcite

Image courtesy of the USGS



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Different Colors of Fluorite

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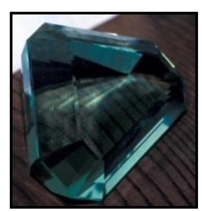

Image courtesy of the Albert Copley Oklahoma University Archives

Different Colors of Quartz

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- 1. Color – every mineral is some color and some are found in multiple colors
- → could be very helpful and distinctive, or could be very ambiguous

Cleavage

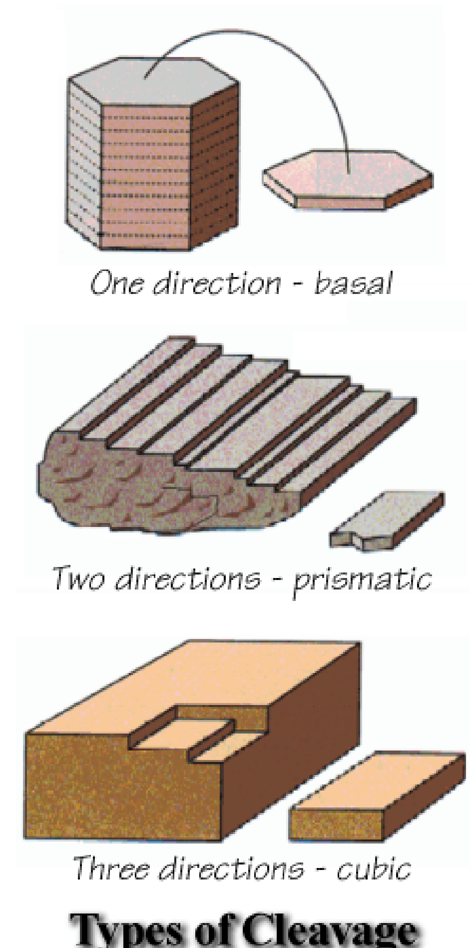
- Cleavage refers to the tendency of a mineral to break along planes of weakness in the chemical bonds, or along planes where bond strength is the least.

- Some minerals break along one dominant plane of cleavage producing parallel sheets, where as others may break along two or more planes of cleavage, producing blocks or prism shapes.

- Not all minerals have distinct planes of weakness that produce cleavage, but those minerals that do, will consistently produce predictable cleavage planes.

Mineral Cleavage

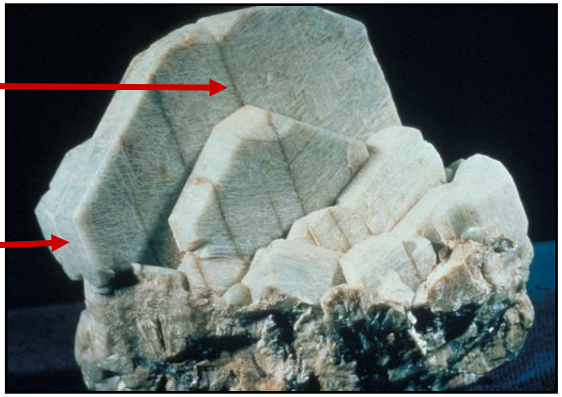
→ Can have multiple planes of cleavage



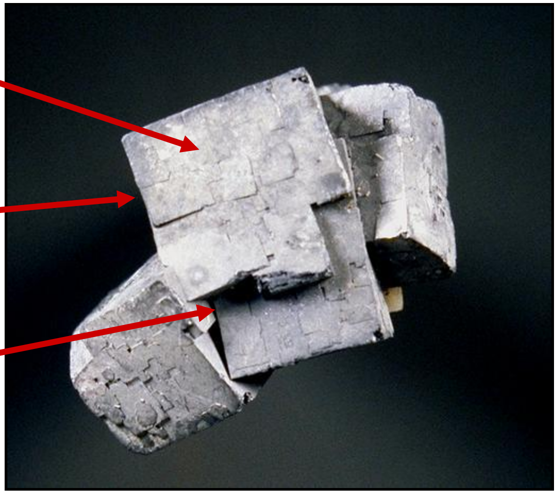
Cleavage, cont.

- One direction of cleavage (one plane)
  - Mineral Example: Micas (muscovite)
- Two directions of cleavage (two planes)
  - Mineral Example: Feldspar
- Three directions of cleavage (three planes)
  - Cubic : Mineral Example: Galena
  - Rhombohedral: Mineral Example: Calcite
- Four directions of cleavage (four planes)
  - Mineral Example: Flourite

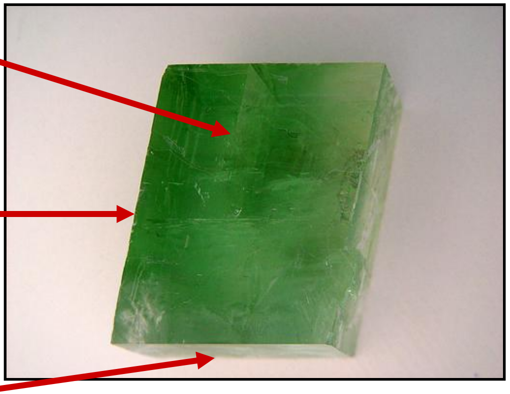
Feldspar: Two Cleavage Planes



Galena: Three Cleavage Planes



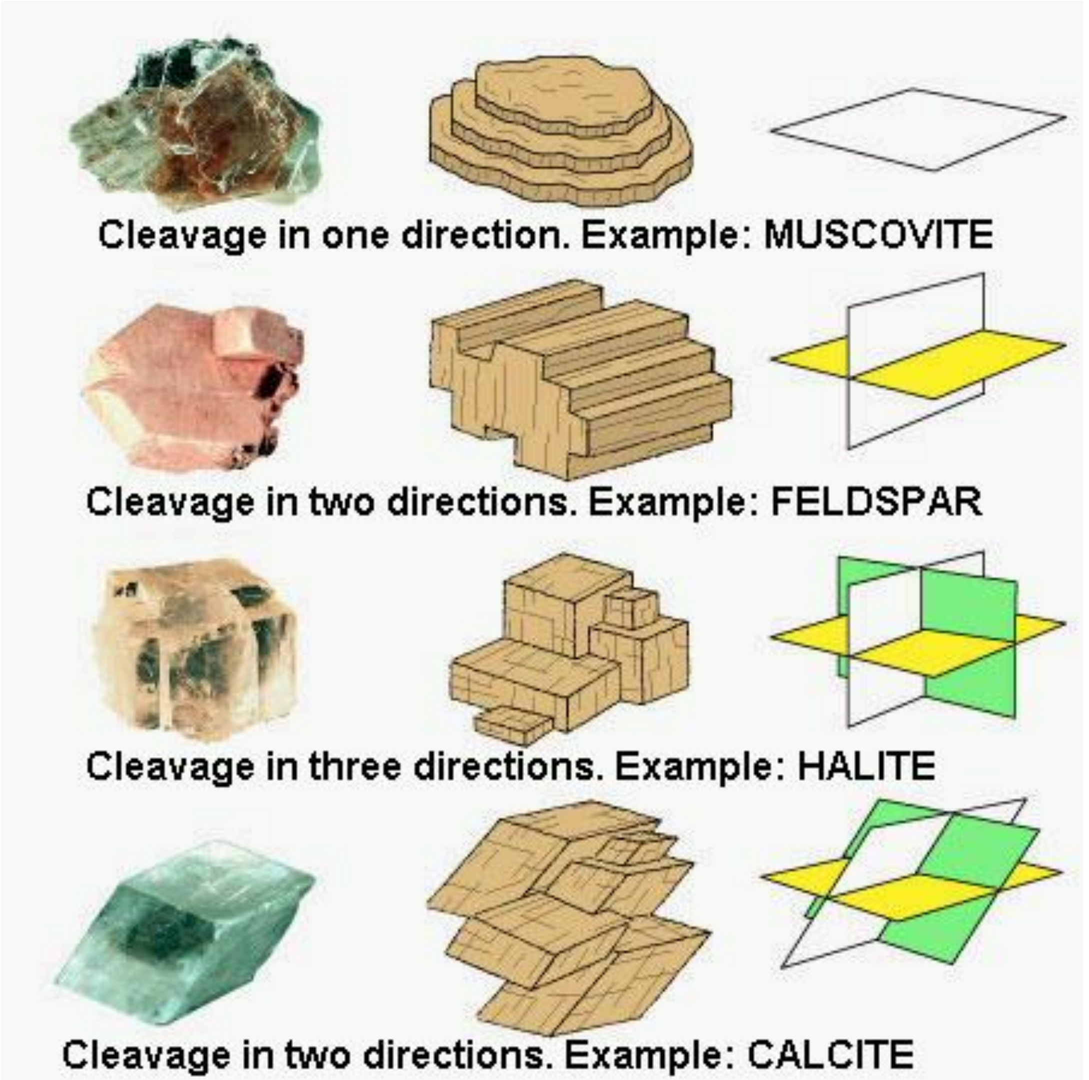
Calcite: Three Cleavage Planes



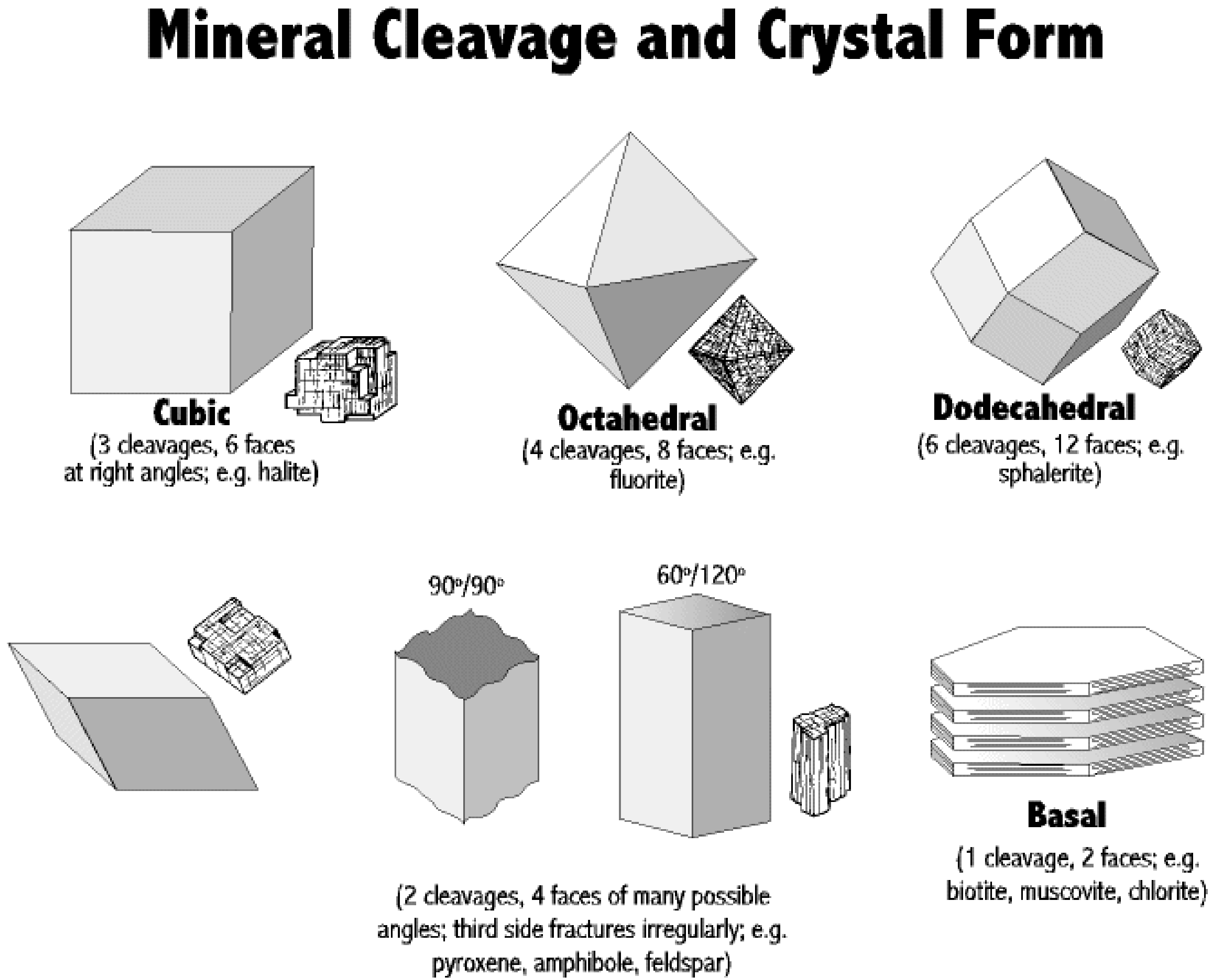
6. Mineral Cleavage – the ability of a mineral to break, when struck along specific planes

→Based on the bonding between atoms

→Where the bonds are weakest = breakage plane



Mineral Cleavage



Mineral Cleavage

→Can have no cleavage (example = quartz)

Fracture

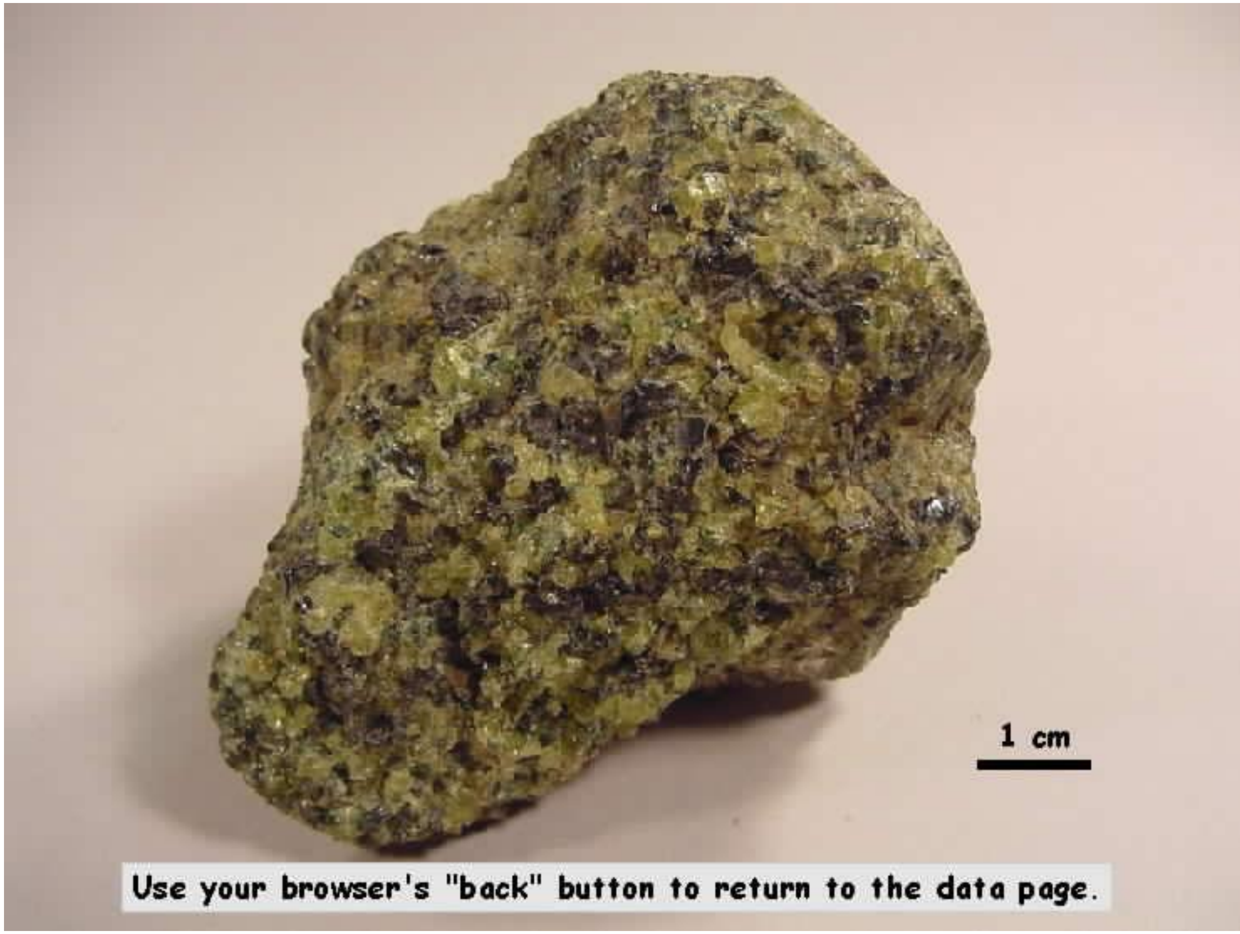
- Fracture refers to the non-planar breakage of minerals.
- Minerals that break along fractures (as oppose to cleavage planes) do not exhibit predictable weakness along specified bonds.
- Fractures may be described as splintery, uneven, or conchoidal.

Conchoidal Fractures on a Quartz Mineral

Fracture

The way a substance breaks where not controlled by cleavage

→Minerals with no cleavage generally break with irregular fracture



Fracture

→ If minerals break with curved fracture surfaces, it is called concoidal fracture

- This is seen in glass, the igneous rock Obsidian,

and the mineral Quartz

Specific Gravity

- Specific gravity refers to the weight or heaviness of a mineral, and it is expressed as the ratio of the mineral's weight to an equal volume of water.
- Water has a specific gravity of 1. Therefore, a mineral with a specific gravity of 1.5, is one and a half times heavier than water.
- Minerals with a specific gravity ≤ 2 are considered light, 2-4 are average, and >4.5 are heavy
- Specific gravity can be measured using complex lab tools such as the hydrostatic balance or more simple procedures involving beakers and water displacement measurements.

Specific Gravity – the density of a mineral

- Density = mass of an object / volume of the object
- The ratio of the mass of an object to the mass of an equal volume of water
- The density of pure water = 1 g / mL
- If the density of the object is < 1 = lighter than water, and will float to some degree
- If the density of the object is > 1 = heavier than water, and will sink
    - Examples:
        - → Quartz = 2.65 g / mL
        - → Galena = 7.5 g / mL
        - → Gold = 19.3 g / mL

Other Special Properties

a. Taste – a few minerals have a characteristic taste

Halite tastes like salt

b. Odor – a few minerals have a characteristic odor

Clay minerals have an “earthy” smell

Physical properties of some non-silicate minerals

Mineral	Formula	Colour	Specific gravity	Hardness	Cleavages
barite	$BaSO_4$	white	4.5	31/2	two
fluorite	CaF_2	variable	3	4	four (parallel to octahedron faces)
Chemical precipitates					
gypsum	$CaSO_4.2H_2O$	colourless or white	2	2	one perfect
halite (common salt)	NaCl	colourless or variable	2	2	three perfect (parallel to cube faces)
Carbonates					
calcite	$CaCO_3$	white or colourless	2.7	3	three (parallel to rhombohedron faces)
dolomite	$CaMg(CO_3)_2$	off-white	3	4	three (parallel to rhombohedron faces)

Physical properties of some ore minerals

Mineral	Formula	Colour	Streak	Specific gravity	Hardness	Cleavages
Sulphides						
galena	PbS	grey	black	7.5	2½	three (parallel to cube faces)
blende (sphalerite)	ZnS	dark reddish brown		4	4	three
pyrite	FeS_2	yellow	black	5	6	none
Oxides						
magnetite	Fe_3O_4	black	black	5+	6±	poor fractures
haematite	Fe_2O_3	brown	brown	5+	5–6	none
limonite	$Fe_2O_3.3H_2O$	earthy	brown	4	5	poor fractures
goethite	$FeO.OH$	yellow	yellow	4	5+	two (prismatic)

Physical properties of some silicate minerals

Mineral	Colour	Specific gravity	Hardness	Cleavages
olivine	green or dark green	3.5+	6½	none (one poor
pyroxene (augite)	black or brown	3.3	5½	fracture) two
hornblende	black	3.3	5½	two
biotite	brown	3	2½	one (perfect)
garnet	red (variable)	3.5+	7	none

Group	Member	Formula	Economic Use
Oxides	Hematite	Fe_2O_3	Ore of iron, pigment
	Magnetite	Fe_3O_4	Ore of iron
	Corundum	Al_2O_3	Gemstone, abrasive
	Ice	H_2O	Solid form of water
	Chromite	$FeCr_2O_4$	Ore of chromium
	Ilmenite	$FeTiO_3$	Ore of titanium
Sulfides	Galena	PbS	Ore of lead
	Sphalerite	ZnS	Ore of zinc
	Pyrite	FeS_2	Sulfuric acid production
	Chalcopyrite	$CuFeS_2$	Ore of copper
	Bornite	Cu_5FeS_2	Ore of copper
	Cinnabar	HgS	Ore of mercury
Sulfates	Gypsum	$CaSO_4 \cdot 2H_2O$	Plaster
	Anhydrite	$CaSo_4$	Plaster
	Barite	$BaSO_4$	Drilling mud
Native elements	Gold	Au	Trade, jewelry
	Copper	Cu	Electrical conductor
	Diamond	C	Gemstone, abrasive
	Sulfur	S	Sulfa drugs, chemicals
	Graphite	C	Pencil lead, dry lubricant
	Silver	Ag	Jewelry, photography
	Platinum	Pt	Catalyst
Halides	Halite	$NaCl$	Common salt
	Fluorite	CaF_2	Used in steelmaking
	Sylvite	KCl	Fertilizer
Carbonates	Calcite	$CaCO_3$	Portland cement, lime
	Dolomite	$CaMg(CO_3)_2$	Portland cement, lime
	Malachite	$Cu_2(OH)_2CO_3$	Gemstone
	Azurite	$Cu_3(OH)_2(CO_3)_2$	Gemstone
Hydroxides	Limonite	$FeO(OH) \cdot nH_2O$	Ore of iron, pigments
	Bauxite	$Al(OH)_3 \cdot nH_2O$	Ore of aluminum
Phosphates	Apatite	$Ca_5(F,Cl,OH)(PO_4)_3$	Fertilizer
	Turquoise	$CuAl_6(PO_4)_4(OH)_8$	Gemstone